

Beating the Clock:

A Predictive Approach to First Responder Response Time

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Abstract:

During emergency situations, first responder response times are crucial. Many factors are uncontrollable; this analysis sought to focus on the attributes that can be influenced, such as inter-agency communication. Using an open-source dataset from NYC, variables were modeled using multiple linear regression (MLR). After a variety of transformations, the MLR did not pass assumptions. Further exploration led to the generation of a more relaxed model – a generalized linear model (GLM) with gamma distribution. LASSO and Ridge regressions were modeled to serve as a comparison to the GLM. Upon reviewing the evaluation metrics, the LASSO model was deemed the best fit.

Introduction:

Emergency situations are some of the scariest, most intense predicaments one can find themselves in. While time for those involved in the incident seems to slow down, the speed at which emergency services (e.g., police, fire, EMS) arrive is more crucial than ever. Wiklund, et. al. (2017), says that “fires can double in size in just 30 seconds.” Additionally, Ibrhaim (2007) states that “for every minute CPR is delayed, the survivability of the injured decreases by 7-10%.” Every second counts, and while emergency services do all they can to arrive on scene as quickly as possible, there are many factors that impede that ability. Variables such as call taker processing time, average distance traveled, and median dispatch time could have a meaningful effect on the arrival time of emergency services. The goal of this study was to determine which factors, if any, have a significant impact on the arrival time of emergency services. As a result of these findings, emergency services would have a better understanding of contributing factors regarding arrival time in hopes of decreasing the time to arrival on the scene.

The original dataset contains one date variable, two categorical variables, and twenty-seven numeric variables. All the numeric variables were recorded in seconds except for `X.of.Incidents.Calculated` which counts the number of incidents in a given month. The two categorical variables are `agency`, which states the department of the emergency service, and `Final.Incident.Type`, which states the final incident category. Twelve numeric variables were department specific (e.g., fire department) measurements. Since the goal of this model was to predict emergency services as a whole, these variables were removed. In addition, the variable of travel time was removed as well, as this is not a factor that can be controlled. In other words, the distance the incident is from the department, the traffic, or the roads traveled cannot be manipulated. It is also understood that when investigating the time to arrival, that the time it takes to travel would be the most dominating factor.

A few of the remaining numeric columns had a singular missing value which was imputed with the mean. A unique ID variable was created for each observation, as well. From the date variable, the month was extracted and sorted into categories by season (three months for spring, summer, fall, and winter respectively) to obtain another categorical variable. Finally, the missing values in Final.Incident.Type were imputed with the mode, and the 23 categories in this variable were condensed down to 10 categories into a new categorical variable called IncidentClassification.

Model:

The data set was split into training and testing data sets using an 80/20 split. The testing data was set to the side, and the training data was utilized for the following steps. To begin the exploratory data analysis (EDA) process, a histogram of the target variable, Call.to.Agency.Arrival, was generated (see Figure 1). The target variable was skewed right, indicating a need for a transformation. Because of the positive skewness, a log transformation was applied (See Figure 2). Histograms were conducted for all numeric response variables, resulting in the same outcome and decision. A correlation matrix was then created to determine the strength of relationship between the variables. It was reported that multiple variables had a correlation coefficient of greater than |0.5|; therefore, the next step was to fit a multiple linear regression (MLR) model.

For the categorical variables, stratified boxplots were created using the variable levels and the log transformed target variable to determine if differences by level occurred (See Figures 3-5). Agency and IncidentClassification showed visual differences; analysis of variance (ANOVA) tests were conducted to confirm statistical significance between the levels. Tukey's post-hoc tests were then run to determine which groups were statistically different from one another. This information was used when building the final model.

Boxplots were also created for the numeric variables to visually assess the presence of extreme outliers (See Figures 6-7). To account for this, the process of winsorization was used, replacing any number below the defined lower limit with the limit itself, and similarly applied to the upper limit.

Once the decision was made regarding which variables to include in the model, an MLR was fitted. Assumptions were checked as the models were created. The MLR model violated the assumptions of normality and homoscedasticity. Due to the skewed nature of the data, the log transformed data was used to create a new MLR (See Figures 8-9). This model also violated the assumptions. Other transformations such as BoxCox and Winsorization were used to build MLR models, but unfortunately none of these models were able to pass the MLR assumptions.

Multicollinearity was then assessed by measuring the Variance Inflation Factor (VIF). Variables were removed one by one until all the variables remaining had a VIF under 10. Again,

the MLR did not pass the assumptions. Therefore, since the data is positively skewed and continuous, a more relaxed model was required.

A generalized linear model (GLM) was generated with a gamma distribution, which is used for right skewed continuous data, and with a log link. One reason the original MLR models created violated assumptions was due to some errors in variance. Additionally, a GLM with gamma distribution, according to Barrett (2019), “does not require integers and can handle more dispersion.” It is used for positive values, as well. All of these descriptors fit the target variable. To ensure best practices, two other models, a Ridge and a LASSO regression, were created. Cross validation was used to determine the best lambda value for the Ridge and LASSO regression models. These models were then compared to the GLM.

Analysis:

The assumptions for the GLM are more relaxed than for a strict MLR model. Additionally, Ghasemi (2012) states that, “For large sample sizes, significant results would be derived even in the case of a small deviation from normality, although this small deviation will not affect the results of a parametric test.” As sample size increases, the assumptions become less crucial.

The two tests used for assumption checking for these models were the Bruesch-Pagan (BP) test and the Kolmogorov-Smirnov (KS) test. Both of these tests are very sensitive to extreme values. So, while the GLM did not pass these assumptions, using Figures 10 & 11 to visually represent the homoscedasticity and normality of the data, respectively, the graphs appear to generally follow the preferred trends.

Since the data had variables that were highly correlated, the LASSO and Ridge regression techniques were important to consider. Those techniques help minimize unimportant variables. They do this by regularizing the coefficients, and LASSO can even effectively remove unimpactful variables by reducing coefficients to 0.

Once all three models were built on the training data, the models were then used to predict values using the test data. The evaluation metrics compared were the Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and Root Mean Squared Error (RMSE). The LASSO regression model had the lowest AIC, BIC, and RMSE values, which points to it being the best fitting model. The RMSE shows an over 2 minute reduction in the error for the LASSO compared to the GLM.

Model	AIC	BIC	RMSE (in sec)	Optimal Lambda
GLM	124006.7	124158.4	795.52	N/A

Ridge	111835.3	111927.0	685.80	120.96
LASSO	111661.6	111739.3	673.70	40.87

To visualize these findings, predicted versus actual plots were created for the GLM and LASSO models (see Figures 12 & 14). The scatterplots show similar trends. With lower arrival times, the accuracy is higher. However, as arrival times increase, accuracy declines due to the heightened influence of factors that cannot be controlled (e.g. distance from emergency services). Finally, to answer the question of which factors are most important, feature importance bar graphs (see Figures 15-16) were generated to visualize which variables have the most effect on predicting arrival time.

Conclusions:

During emergency situations, there are many things out of one's control. Circumstances such as the reaction time for someone to actually call 911, how far away a station is from the location of the emergency, or the severity of the incident are not components that can be controlled or improved upon. However, the internal processing time once the call has been made *is* something that can be investigated for meaningful results. The amount of time it takes between transferring calls, the response time per department, and the urgency as a result of the assumed incident are factors that can be optimized to decrease the amount of time needed to get to the scene of the emergency.

Upon comparing the three models generated, it was determined that the LASSO model was the best performing. The LASSO model not only had the lowest AIC and BIC values, but also the lowest RMSE value. This indicates that when predicting the arrival time of first responders, the LASSO model produces predictions that are closest to the actual value in comparison to the other models. While on the surface the RMSE values are higher than ideal, with the selected model's RMSE value being 673.7, or roughly 11 minutes, this is not quite as drastic as it may appear. Considering the average arrival time is approximately 28 minutes, with the maximum range being over four hours, an error of 11 minutes seems reasonable. Considering the target variable is the time it takes to arrive at the scene of an emergency, one must keep in perspective that travel time is an inherent component of this.

The goal was to provide meaningful insight to those directly involved in the first responder arrival time, whether that be those answering the phone, transferring the call, or the ones actively going on the scene. Feature importance from the generated LASSO model indicated that the agency assigned to the incident, the time to assign the proper agency, and the time to answer the initial 911 call have the greatest impact on the time to arrival. These are components that have the potential to be improved upon. It is inferred that if these specific variables are enhanced, that the overall time to arrive at the scene of an emergency would decrease.

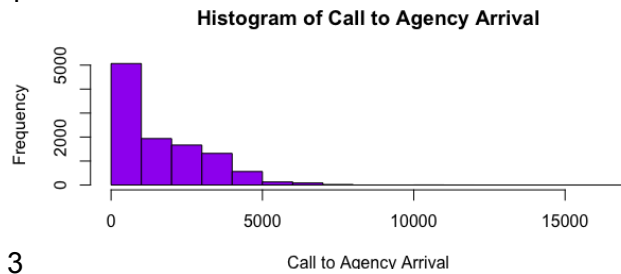
In summary, while the circumstances around emergency situations are uncontrollable and essentially unpredictable, there are conditions that can be improved to increase the overall efficiency of arriving at the scene. Future studies would benefit from investigating factors such as call volume and how the current technology handles overload, location accuracy using Geographic Information Systems, and staffing standards including shift changes and crew size. In addition, utilizing other analyses such as time series, survival, or classification could provide beneficial information to answering these inquiries. The proposed objectives, in combination with currently obtained data, could prove invaluable to greatly improving the speed at which emergency responders are able to arrive on the scene.

References:

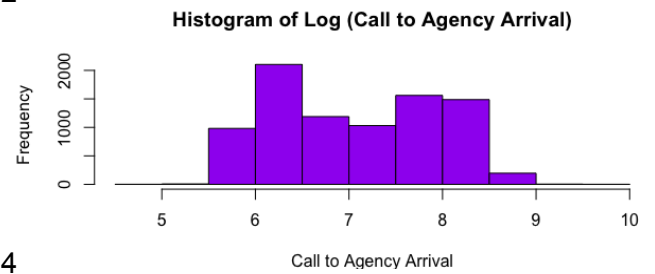
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Appendix

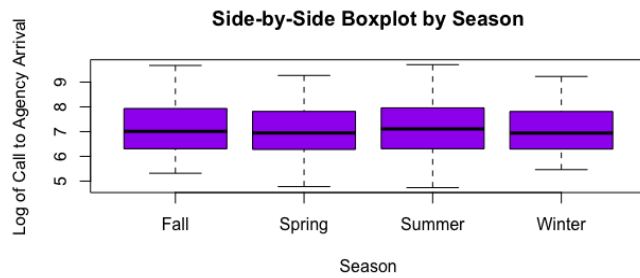
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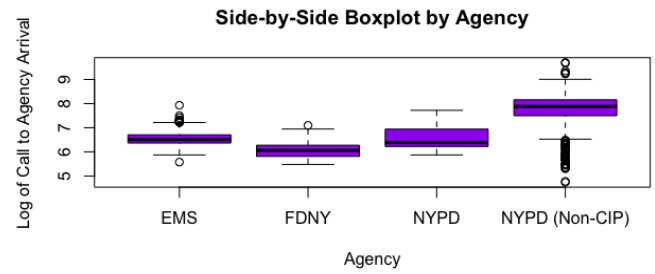
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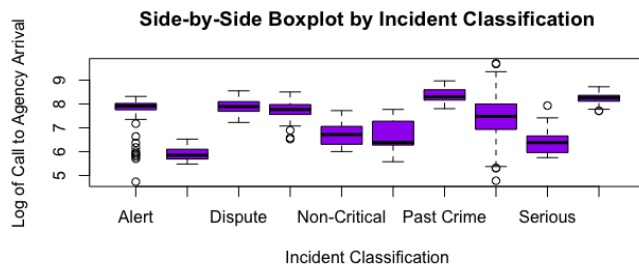
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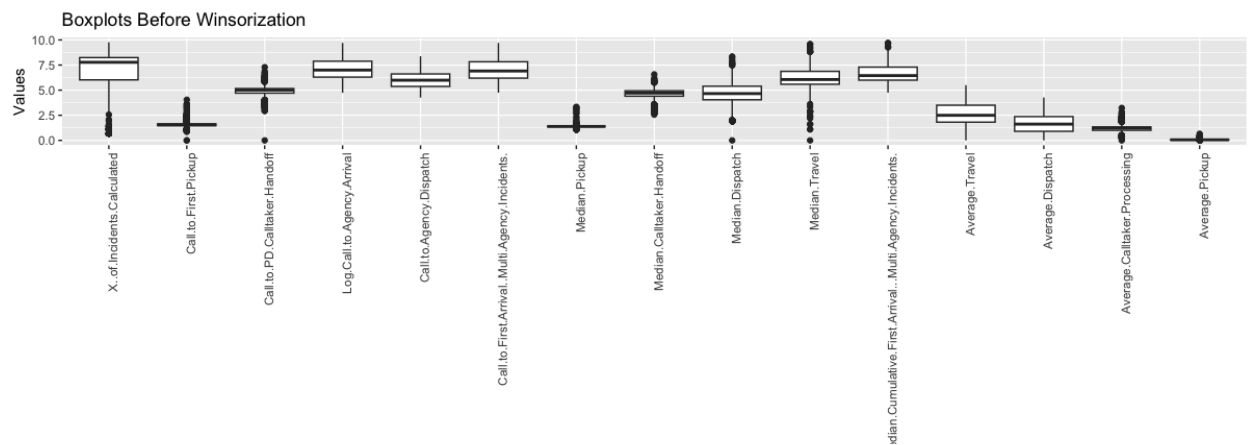
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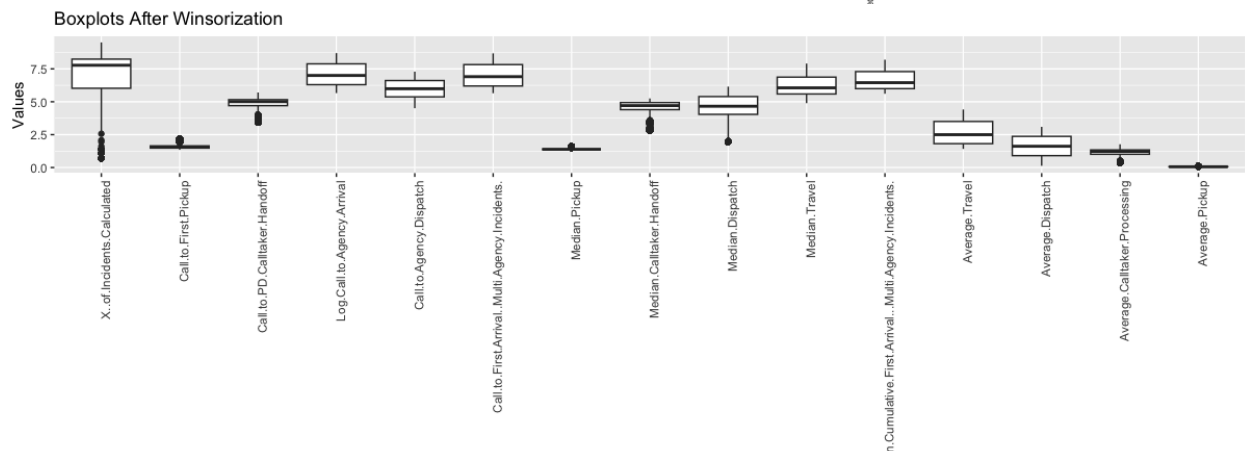
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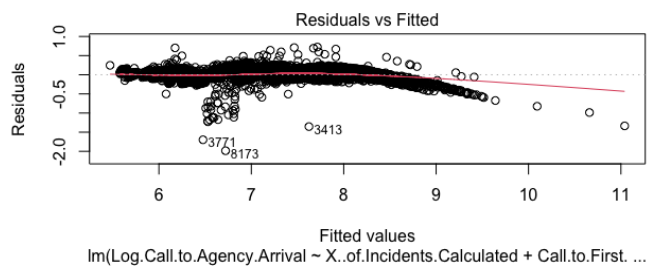
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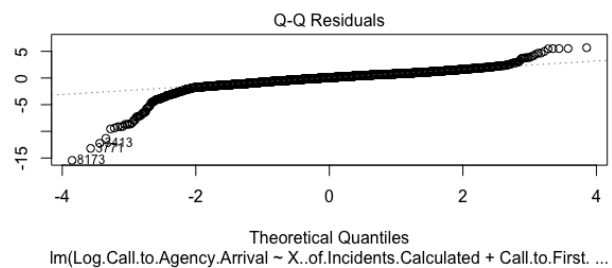
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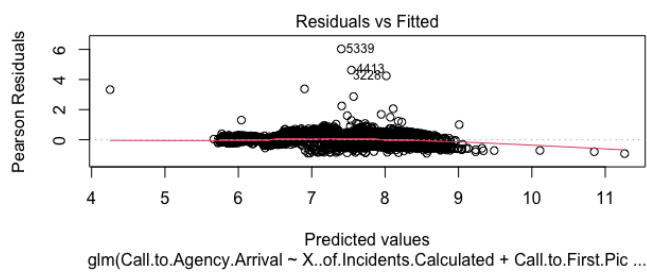
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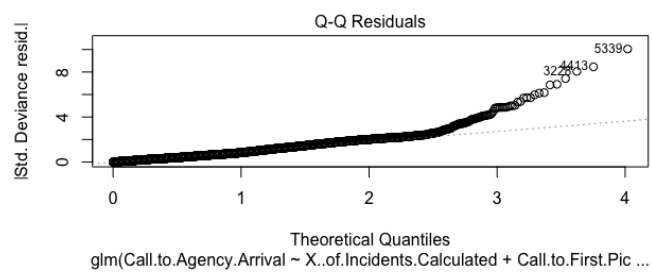
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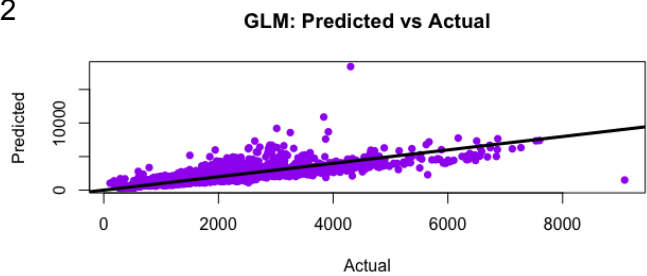
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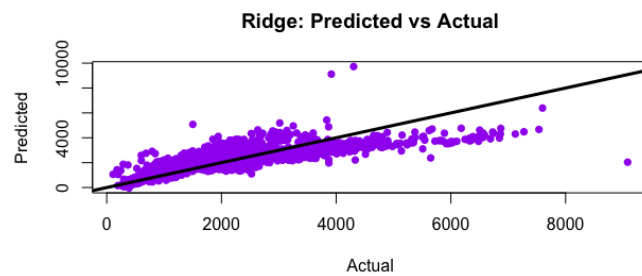
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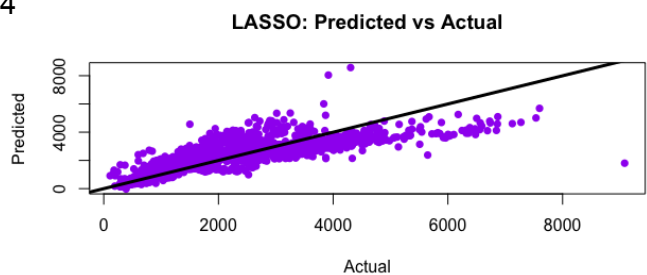
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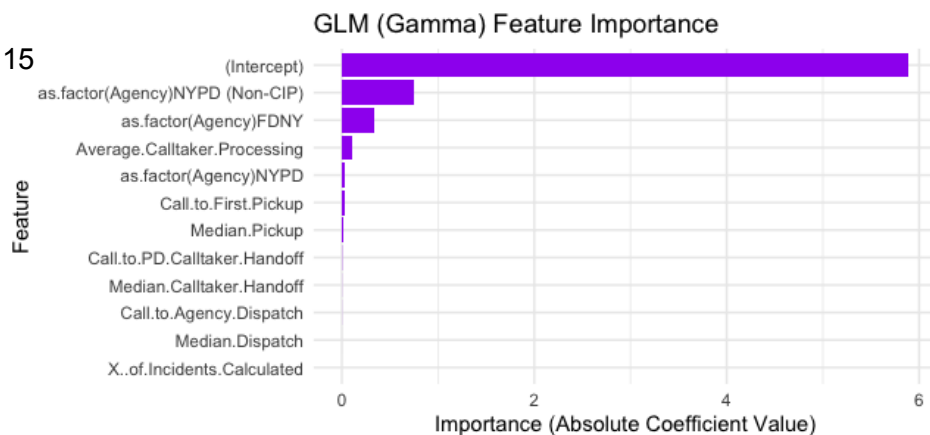
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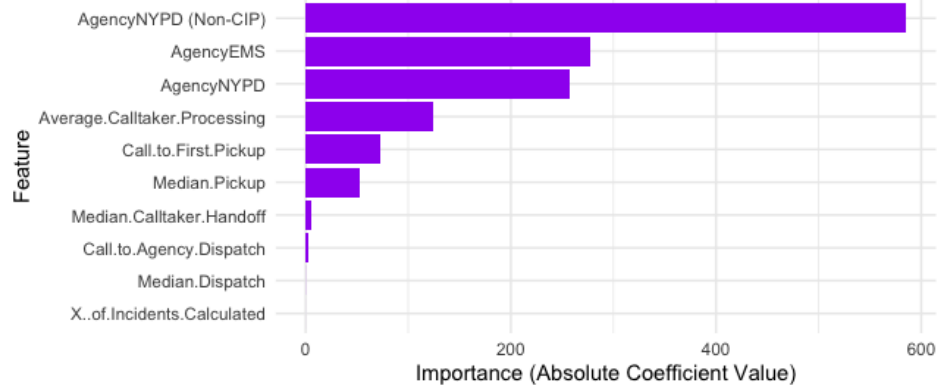
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LASSO Feature Importance



Data Dictionary for First Responder Data

Variable Name	Description	Units/Grouping
ID	Identifier variable	N/A
Week.Start.Date	Week start date of the incident	mm/dd/yyyy
Season	The time of year of the incident	Winter (Dec, Jan, Feb), Spring (Mar, Apr, May), Summer (June, July, Aug), Fall (Sept, Oct, Nov)
Agency	Which emergency service department	FDNY (New York Fire Department), EMS (Emergency Medical Service), NYPD (New York Police Department), NYPD non CIP (non crime in progress)
Final.Incident.Type	Type of emergency incident	Alert (Alarms/Shot Spotter), Critical (Life threatening medical emergencies and structural fires), Dispute (altercations and disorderly conduct), Hazard (Hazardous material scares and USPS bio hazard), Non-Critical (medical emergencies), Other (non-medical emergencies and suspect detainment), Past Crime, Possible Crime (investigation, other crimes), Serious (non-life threatening medical emergencies, non-structural fires), Vehicle Accident
X..of.Incidents.Calculated	Number of Incidents Calculated	Total per month
Call.to.First.Pickup	Time it takes for the 911 call to be answered	Seconds
Call.to.PD.Calltaker.Handoff	Time it takes for the call to be transferred from 911 to correct agency	Seconds
Call.to.Agency.Dispatch	Time it takes for the call to be assigned to the correct agency	Seconds
Call.to.Agency.Arrival	Time from initial call until designated agency arrives on scene	Seconds
Call.to.First.Arrival..Multi.Agency.Indicents	Time from initial call until ANY agency arrives on scene	Seconds
Median.Pickup	Median time it takes for the 911 call to be answered	Seconds
Median.Calltaker.Handoff	Median time it takes for the call to be transferred from 911 to correct agency	Seconds
Median.Dispatch	Median time it takes for the call to be assigned to the correct agency	Seconds
Median.Travel	Median time from initial call until designated agency arrives on scene	Seconds
Median.Cumulative.First.Arrival...Multi.Agency.Incidents.	Median time from initial call until ANY agency arrives on scene	Seconds
Average.Travel	Average time from when the agency was assigned until arriving on the scene	Seconds
Average.Calltaker.Processing	Average time from answering the call to assigning to the designated agency	Seconds
Average.Dispatch	Average time from the transfer until a unit is assigned	Seconds
Average.Pickup	Average time for 911 to answer the initial call	Seconds